Makayla Cheng

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA ***Aug 2023 - May 2027***

B.S.E. in Bioengineering, Minor in Computer Science *GPA*: 3.82/4.0

Accelerated Master's in Robotics (Expected May 2027)

Relevant Coursework: Mechatronics, Statics, Dynamics, Biomaterials, Statistics, Mechanical Design (CAD), Robotics, Python

TECHNICAL SKILLS

Robotics & Controls: Motion Control, Mechatronics, Computer Vision, Machine Learning, Control Systems, Arduino Microcontrollers, Circuit Design

Design & Validation: SolidWorks, AutoCAD, 3D Printing, Rapid Prototyping, Design Verification, Fatigue Testing, DFM

Programming & Modeling: ROS, Gazebo Python, MATLAB, R, Java, OCaml, HTML, SQL, Simulation

PROFESSIONAL EXPERIENCE

Medtronic, *Mechanical Design R&D Intern* - Boston, MA ***June 2026 - Aug 2026***

- Designed and implemented an automated design verification test method for a cardiac ablation catheter, investigating surgeon-reported wire failures and supporting root-cause analysis and product development decisions.
- Engineered and validated a custom electromechanical test fixture integrating SolidWorks-designed components, 3D-printed parts, embedded microcontrollers, and motorized actuation to improve repeatability and reduce testing time.
- Executed verification testing and fatigue characterization across catheter samples to identify failure mechanisms.
- Developed standardized test procedures and documentation to support future R&D validation and NPI readiness.

Yim Lab, *Undergraduate Researcher* - Philadelphia, PA ***August 2025 - Present***

- Designed and iteratively prototyped a novel 3D-steerable surgical needle using CAD and tape-spring mechatronics, achieving sub-millimeter positioning precision for minimally invasive surgical navigation.
- Applied machine learning models to characterize steering behavior and improve trajectory accuracy, reducing trajectory error by 25% in gel-tissue experiments.
- Integrated mechanical design, actuation, modeling, and hands-on testing to advance a robotic navigation concept for minimally invasive procedures.

AstraZeneca, *Pharmacokinetics Modeling Intern* - Boston, MA ***June 2025 - August 2025***

- Independently designed and led development of mathematical models in MATLAB for T-cell engager therapies, directly guiding dose selection and advancing candidates into first-in-human oncology trials.
- Conducted 600+ simulations to optimize binding, enhancing T-cell trimer formation and therapeutic efficacy by 35%.

Airalux, *Technical Intern* - Philadelphia, PA ***January 2025 - June 2025***

- Conducted 100+ interviews with nurses, respiratory therapists, and hospital staff to identify usability gaps, workflow constraints, and unmet clinical needs for a therapeutic spirometry device.
- Translated clinician feedback into actionable engineering requirements and design-for-manufacturing improvements, reducing out-of-spec assemblies by 80%.
- Bridged clinical needs with engineering execution through prototype verification and design iteration.

Cerespectus (Penn Innovation Startup), *Engineering Intern* - Philadelphia, PA ***May 2024 - Present***

- Assisted in preparation of FDA design control documentation, risk management files, and engineering documentation for a Class II CNS delivery catheter.
- Gained working knowledge of medical device regulatory requirements and design verification processes.

PROJECTS

Cockroach Machine Interface ***October 2025***

- Designed an electromechanical test platform integrating servo motors, EMG electrodes, and biological tissue.
- Developed MATLAB control software for PWM motor control and automated experimental testing.

Pfizer Hackathon (First Place Winner) ***August - September 2025***

- Led end-to-end development of an agentic AI–powered solution, designing and implementing full-stack web architecture (front-end, back-end, database integration, and automated testing)
- Leveraged ML techniques to extend Pfizer's existing digital platforms for climate change associated risks.